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SUPERSEDING
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FEDERAL SPECIFICATION

WIPING CLOTHS, COTTON

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—Cotton wiping cloths covered by this specification are primarily intended for use in wiping machinery and paint, and for miscellaneous cleaning.

1.2 Classification.

1.2.1 *Type and classes.* — Cotton wiping cloths covered by this specification shall be of one type and the following classes, as specified:

Class 1.—Unused fabrics
Class 2.—Reclaimed fabrics (see 3.1.2)

1.2.2 *Colors.*—The wiping cloths shall be furnished in the following colors, as specified:

Undyed
Mixed colors (which may include undyed cloths)

2. APPLICABLE SPECIFICATIONS

2.1 The following Federal Specification, of the issue in effect on date of invitation for bids, forms a part of this specification:

CCC–T–191—Textile Test Methods.

(Copies of Federal specifications and the Federal Specifications Index may be obtained upon application, accompanied by money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C., or to the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Seattle. Prices may be obtained from the Superintendent of Documents or from the GSA Regional Offices.)

2.2 Specifications and other publications applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 Material.

3.1.1 *Class 1.*—Class 1 wiping cloths shall be of soft, absorbent cotton fabrics, free from stain, dust, abrasive material, and alkali. The wiping cloths shall consist of clean mill ends, mill remnants, or both. Terry woven fabrics, heavily napped fabrics, mesh fabrics having openings larger than ⅛-inch square, or weighing less than 3 ounces per square yard, fabrics woven with hard twisted yarns, and starched or stiffened fabrics, will not be acceptable.

3.1.2 *Class 2.*—Class 2 wiping cloths shall be soft, absorbent cotton fabrics consisting principally of fabrics reclaimed from household articles and garments. Wiping cloths having the following defects will not be acceptable:

(a) Paint and unsightly or excessive oil stains which retard absorbency (see 3.4).

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(b) Cloths containing dust, abrasives, or alkali.

(c) Cloths dyed to obliterate stains or other defects.

(d) Cloths made of mesh fabric having an opening larger than ⅛ inch square or made of mesh fabric weighing less than 3 ounces per square yard.

(e) Cloths so badly worn or tendered as to be unsuitable for the use intended, cloths having stray ends narrower than 3 inches and protruding from the body of the cloth more than 6 inches, or tattered ends more than 2 inches long.

(f) Cloths made of unopened sleeves, trouser legs, and tubular fabrics and unopened hems over 1 inch wide and trouser cuffs on fabrics weighing more than 6 ounces per square yard. Pockets, reinforcements (including collars, cuffs, button hole and waist band reinforcements) and patches on fabrics over 6 ounces per square yard.

(g) Cloths containing straps, buttons, hooks, eyes, pins, elastic bands, and any other metallic fittings.

(h) Cloths made of U. S. flags, flags of other nations, or remnants thereof.

3.2 Size.

3.2.1 *Class 1.* — Each wiping cloth shall have an area of not less than 2 square feet, and be not less than 9 inches wide and not more than 44 inches long. Cloths which have been sewn together to obtain the minimum size will be accepted provided not more than two pieces of fabric have been used.

3.2.2 *Class 2.* — Each wiping cloth shall have an area of not less than 2 square feet, and be not less than 9 inches wide and not more than 44 inches long, except that it may be not less than 6 inches wide for not more than 6 inches in length. Cloths which have been sewn together to obtain the minimum size will be acceptable provided not more than two pieces of fabric have been used.

3.3 Weight.

3.3.1 The wiping cloths shall weigh not more than 14 ounces per square yard nor less than 2 ounces per square yard. At least 65 percent of the weight of the cloths in each bale shall weigh between 2 and 9 ounces per square yard.

3.3.2 When specified, the wiping cloths shall weigh not more than 9 ounces per square yard, nor less than 3 ounces per square yard. At least 65 percent of the weight of the cloths in each bale shall weigh between 3 and 6 ounces per square yard.

3.4 Absorbency. — Cotton wiping cloths shall be absorbent, and when tested in accordance with 4.3.2, the cloths shall be considered absorbent if the water and oil are absorbed into the fabric within 30 seconds. However, if the water or oil is not absorbed but remains wholly or partly as a film on the surface of the fabric or on the supporting surface, the cloth shall be considered non-absorbent.

3.5 Washing and sterilization.

3.5.1 *Class 1.*—Class 1 wiping cloths may be washed.

3.5.2 *Class 2.*—Class 2 wiping cloths shall be washed and sterilized (this processing shall be done within the continental limits of the United States). (See 6.1(a).)

3.6 Basis of purchase. — Wiping cloths shall be purchased by net weight provided the moisture content as determined in 4.3.1 is 6.5 ± 1 percent (based on commercial moisture regain of 7 percent as covered in Commercial Standard, Moisture Regains of Cotton Yarns, CS 11–41). Deliveries containing less than 5.5 and more than 7.5 percent moisture shall be purchased by net weight adjusted to a basis of cloths containing 6.5 percent moisture (see 6.2). Cloths shall not have a moisture content greater than 10 percent.

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3.7 **Workmanship.** — The wiping cloths shall be clean, dry, and free from objectionable odor.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.

4.1.1 Unless otherwise specified, one percent of the number of bales in a delivery shall be taken at random by the inspector for inspection and tests, except that in no instance shall less than three bales be taken.

4.1.2 From each of the bales taken, the inspector shall take three samples, each containing approximately 1 pound of cloths. Two of the samples shall be taken from opposite sides of the bale and the third from near the center. If scales are available at the time of sampling, the samples shall be weighed immediately after removal from the bales. If scales are not available, the samples shall be placed immediately in air-tight containers, sealed, and forwarded to the laboratory for moisture content determinations.

4.2 Inspection.

4.2.1 *Preliminary.* — The inspector shall thoroughly inspect the cloths in the selected bales for quality, cleanliness, size, color, stains, weight, ragged and worn parts, buttons and other attachments, dirt, and seams. Cloths which are deficient in any respect will be considered unsatisfactory. (If the weight of unsatisfactory materials is more than 7½ percent of the weight of the cloths in the sample bales, the entire lot submitted for inspection shall be rejected.)

4.2.2 *Final.*—If the cloths in the sample bales meet the requirements of this specification, the inspector may, if he deems it advisable, inspect additional bales in such a manner as may be necessary to insure that the quality of the entire lot is equal to that of the bales examined in accordance with 4.2.1. (If the weight of unsatisfactory materials is more than 7½ percent of the weight of the cloths in the sample bales, the entire lot submitted for inspection shall be rejected.)

4.3 **Tests.**—Tests shall be conducted in accordance with the applicable methods described in Federal Specification CCC–T–191, and as hereinafter specified.

4.3.1 *Moisture content.*—The samples for test shall be accurately weighed and then dried in an oven at 221° to 230° F. (105° to 110° C.) until the weights from two consecutive weighings, taken not less than 10 minutes apart, differ by not more than 0.1 percent of the first of the two weights. The moisture content is the difference between the weight of the cloths as sampled (Wo) and their "dry weight" (Wd), expressed as a percentage of the original (sampled) weight:

$$\text{Moisture content (percent)} = \frac{Wo - Wd}{Wo} \times 100$$

The average of the moisture contents of the individual samples shall be considered the moisture content of the entire lot represented by the samples.

4.3.2 *Absorption.*

4.3.2.1 The sample cloth shall be laid on a nonabsorbent surface and several drops of water and a light machine oil applied. After 30 seconds the sample shall be examined and the absorption or nonabsorption of the water and oil reported.

4.3.2.2 In cases of dispute, drops 0.04 milliliter in volume shall be applied to the sample cloths through a capillary by allowing the drops to fall freely into the material from a height of ½ to 2 inches.

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5. PREPARATION FOR DELIVERY

5.1 Baling.

5.1.1 *Weight and cubic content of bales.*— The bales shall conform to the following requirements, as specified.

Net weight in pounds	*Volume in cubic feet*
45 to 55	2½ to 3½
90 to 110	5 to 7
540 to 660	25 to 35

5.1.2 The wiping cloths shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate and to afford maximum protection from normal hazards of transportation.

5.2 **Marking.**—Unless otherwise specified, each bale shall be marked with the name and class of the commodity as defined by the contract or order under which the shipment is made, the name of the contractor, the number of the contract or order, the name and address of the consignee, the net weight of the wiping cloths, the tare, and the gross weight.

6. NOTES

6.1 **Ordering data.** — Purchasers should specify the class, color, and weight range of the cloths required and the size of the bales to be delivered (see 1.2, 3.3.2, 4.1.1 and 5.1.1). (The larger bales conform to general commercial practice and should be purchased whenever practical.) Purchasers should also exercise any desired options offered herein (see 5.2) and should specify the following contractual requirements:

(a) That a certified copy of any affidavit, by a responsible officer of the contractor's organization, be submitted with each shipment to the effect that the cloths furnished have been washed, sterilized, and dried within the continental limits of the United States. (See 3.5.2.)

(b) That net invoice weights governing payment will be on the basis of net weight of deliveries. (See 3.6.)

6.2 **Adjustment of net weight.** — Where the moisture content as determined in 4.3.1 is not within the limits specified in 3.6 the net weight may be adjusted by the formula:

$$W = \frac{N \times (100 - C)}{93.5}$$

where:

W = adjusted net weight.
N = net weight as delivered.
C = moisture content, in percent (4.3.1).

The moisture content specified in 3.6 is an approximation based on a commercial moisture regain of 7 percent as defined by Commercial Standard Moisture Regains of Cotton Yarn, CS 11–41.

6.3 Transportation description.

6.3.1 *Transportation description applicable to these items is:*

Cloths or rags, wiping.
*In machine pressed bales or machine pressed in boxes.
In packages not machine pressed.
Carload weight 24,000 pounds, subject to Rule 34, Consolidated Freight Classification.
Truckload weight 24,000 pounds.

*Preferred method of shipment to obtain lowest transportation charge.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definite related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following departmental specifications and standards, of the issues in effect on date of invitation for bids, and special

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requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army, Navy, and Air Force.

7.2.1 *Applicable specifications and standards.*

Federal Specification:

QQ–S–790—Strapping, Round; Steel, Bare and Zinc-Coated.[1]

Military Specification:

MIL–B–3149—Baling of Clothing and Equipage.[1]

Navy Department Specification:

General Specifications for Inspection of Material.[1]

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.[1]

MIL–STD–129—Marking of Shipments.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.2 *Sampling, inspection, and test procedures (Navy only).*

7.2.2.1 *Inspection procedures.*—General inspection procedures shall be in accordance with General Specifications for Inspection of Material.

7.2.2.2 *Sampling.*

7.2.2.2.1 *Lot.*—For purposes of sampling, a lot shall consist of not more than one carload of wiping cloths of one class and color offered for delivery at one time.

7.2.2.2.2 *Sampling for visual inspection.*—Sampling for visual inspection shall be in accordance with Military Standard MIL–STD–105. The Acceptable Quality Level shall be 4 percent. An item of product shall be 1 pound of cloths. A sufficient number of bales shall be taken at random by the Naval inspector to make up the required sample poundage.

7.2.2.2.3 *Sampling for test for moisture content.*—From the lot, three bales shall be taken at random, opened, and three samples of 1 pound each (approximately) taken from each bale. Two of the three samples shall be taken from opposite sides of the bale and the third from near the center. The nine samples shall be kept separated so that nine determinations of moisture content can be made. If equipment is not available in the contractor's plant for moisture determination, the nine samples shall be placed in airtight containers as soon as possible and forwarded to the laboratory designated by the procuring agency concerned for moisture content determinations.

7.2.2.3 *Inspection.*

7.2.2.3.1 *Weight.*—To determine conformance with 3.3, each of the sample bales taken in accordance with 7.2.2.2.3 shall be sorted into three piles, as follows:

(a) Acceptable cloths weighing 2 to 9 ounces per square yard.

(b) Acceptable cloths weighing 9 to 14 ounces per square yard.

(c) Nonacceptable cloths of all classes (for inspection under 7.2.2.3.2).

7.2.2.3.2 *Visual inspection.*—The sample shall be examined for conformance with section 3. When the absorbency of a cloth is in doubt, the methods described in 4.3.2 shall be followed by the Naval inspector. Nonconforming cloths in each selected bale shall be set aside and weighed. If the total weight of nonconforming cloths in the sample exceeds the applicable acceptance number (in pounds), the entire lot shall be rejected.

[1] Applicable to Navy purchases only.

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7.2.2.4 *Test for moisture content.* — Test shall be conducted in accordance with 4.3.1, and if the average moisture content of the individual samples exceeds 10 percent, the entire lot shall be rejected.

7.2.3 *Baling (Navy only).*—Unless otherwise specified wiping cloths shall be baled in 45- to 55-pound bales in conformity with the requirements of Military Specification MIL–B–3149, with the following additions and exceptions:

a. Baling boards (3.1.1) will not be required.

b. Either flat strapping as described in paragraph 3.1.4 or class B, zinc-coated (galvanized) round wire strapping conforming to Federal Specification QQ–S–790 is acceptable provided such wire strapping has a minimum diameter of 0.0720 inch.

c. Burlap shall be clean and have a strength closely approximating that of new material but need not necessarily be new.

d. Wire ties (3.1.6) may be 16-gage soft iron or steel with a ½-inch diameter formed eye at each end.

7.2.4 *Marking.*

7.2.4.1 *Army and Navy.*—Each bale shall be marked in accordance with Military Standard MIL–STD–129.

7.2.4.2 *Air Force.* — In addition to the marking required in 5.2 shipping containers shall be marked in accordance with Military Standard MIL–STD–129. The nomenclature shall be as follows: Wiping Cloths, Cotton (*Mixed Colors or Undyed), General Purpose, Federal Specification DDD–W–415a.

*Applicable data to be entered by the contractor.

7.2.5 *Ordering data (Navy only).* — Procurement documents should specify the following:

a. Title, symbol, and date of this specification.
b. Class required (see 1.2.2).

NAVY INTEREST: *MC* Or Sh S Y

7.3 Post Office Department.

7.3.1 *Sampling, inspection, and testing.*—Wiping cloths purchased by the Post Office Department shall be in accordance with the outline contained in the invitation for bids.

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